

Application to make the Gas filling Station easy using CRM (Developer)

Abstract:

This project focuses on the development of a Salesforce CRM application tailored specifically for gas filling stations to modernize and automate their business operations. Traditional gas stations often face challenges such as manual record-keeping, inefficient customer service, and a lack of real-time data for decision-making. By leveraging Salesforce's cloud-based CRM capabilities, this application offers a centralized platform for managing customer relationships, tracking fuel sales, monitoring inventory levels, scheduling deliveries, and handling billing operations.

The solution integrates key Salesforce features such as Service Cloud, Sales Cloud, and custom Lightning components to offer a user-friendly, scalable, and mobile-accessible interface. With automated alerts, detailed dashboards, and real-time analytics, station operators can enhance customer satisfaction, optimize stock management, and drive sales growth. This application serves as a digital transformation tool for fuel station owners, bringing operational efficiency and data-driven management into a traditionally manual industry.

Objective:

The primary objective of this project is to develop a Salesforce-based CRM application that simplifies and automates the day-to-day operations of gas filling stations. This includes improving customer relationship management, streamlining fuel sales and inventory tracking, enabling real-time reporting, and enhancing service delivery through automation. The solution aims to empower station owners and staff with digital tools for efficient decision-making, better customer engagement, and increased operational efficiency.

- Automate customer data management and fuel transaction tracking.
- Streamline inventory control for fuel stocks and cylinders.
- Enable real-time reporting and performance dashboards.
- Improve customer engagement with CRM features like loyalty programs and service reminders.

- Provide mobile-friendly access for on-site and remote station management.
- Reduce manual errors and enhance overall operational accuracy.
- Ensure scalability and flexibility for future growth and third-party integral

TECHNOLOGY DESCRIPTION:

1. Salesforce

Salesforce is a cloud-based CRM platform used to manage customer data, automate processes, and improve efficiency. [A]

2. Object

Objects in Salesforce are like database tables.

- Standard Objects: Already available (e.g., Account, Contact, Opportunity).
- Custom Objects: Created for project-specific needs (e.g., Gas Booking, Cylinder Delivery, Payment).

Each object holds related data and helps organize the station's business processes.

3. Tabs

Tabs are the front-end representation of objects. They make objects easily accessible to users.

Example: A Gas Booking Tab allows staff to enter quickly and view customer gas booking details.

4. The Lightning App

A Lightning App bundles tabs, dashboards, reports, and utilities in one place for users.

The 'Gas Station Management App' centralizes booking, delivery tracking, and billing, simplifying navigation for employees.

5. Fields

Fields are columns in objects that store specific data.

Example:

- In Gas Booking: Customer Name, Cylinder Type, Booking Date.
 - In Payment: Amount, Payment Method, Transaction ID.
- Custom fields ensure the app collects all required information for smooth operations.

6. Page Layouts

Page layouts organize fields, sections, and related lists on a record page.

For example, the Gas Booking layout shows customer details at the top, booking details in the middle, and payment history at the bottom—organized for easy use.

7. Profiles

Profiles define what users can do in Salesforce (object-level and field-level access).

Example:

- Manager Profile: Full access to all records.
- Staff Profile: Limited to booking and delivery details only.

8. Role & Role Hierarchy

Roles define data visibility based on hierarchy. Higher roles can see data of users below them.

Example:

- Station Manager (top role) → Can see all customer and delivery data.
- Delivery Staff → Can only see their assigned deliveries.

9. Users

Salesforce creates each employee as a user..

Example:

- Delivery Boy (User with Delivery Staff role).
 - Station Manager (User with Manager role).
- Assigning correct profiles and roles ensures proper access.

10. Permission Sets

Permission sets give additional permissions without changing the profile.

For example, a delivery staff member might need temporary access to generate reports, which the team can provide via a Permission Set without modifying their base profile.

11. Setup for OWD

Organization-Wide Defaults defines the baseline data visibility across the organization.

- For this project:
 - Gas Booking: Private (only the owner and managers can see).
 - Payment: Controlled by Parent (visible through related booking).

This ensures data security and confidentiality.

12. User Adoption

User Adoption ensures employees understand and effectively use the system.

- Training sessions for staff.
- Easy navigation with customized Lightning App.
- Reports/Dashboards to encourage data-driven work.
Strong adoption helps the project succeed in real-time.

13. Reports

Reports provide structured insights from the data.

Examples:

- Daily bookings report.
- Payment status report.
- Delivery performance report.

These reports help the management make decisions quickly.

14. Dashboards

Dashboards give a visual representation of reports.

Examples:

- Pie chart for booking types.
- Bar chart for payments collected per week.
- Line chart for delivery trends.
This makes station performance easy to monitor at a glance.

15. Flows

Flows automate business processes with clicks (no code).

Example:

- Auto-send SMS/email confirmation to customers after a gas booking.
- Booking confirmation automatically generates a delivery task.
- Flows reduce manual work and increase efficiency.

16. Apex Trigger

Apex Triggers perform tasks that Flows cannot handle.

Example:

- A trigger reduces cylinder stock in the Inventory object upon Gas Booking delivery.

Triggers ensure accurate, real-time updates across objects.

Project implementation steps:

Creating a Salesforce Developer Account

1. Visit: <https://developer.salesforce.com/signup>
2. Fill the sign-up form with:
 - Name: First & Last Name
 - Email: Valid email address
 - Role: Developer
 - Company: Your College Name
 - Country: India
3. Click Sign Me Up to complete registration.

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First name: Bobbadi ✓ Last name: Hemanth ✓

Job title: Developer ✓ Work email: 22kd1a0213@lendi.edu ✓

Company: Lendi Institute of eng ✓ Country/Region: India ✓

Your org may be provisioned on or migrated to Hyperforce, Salesforce's public cloud infrastructure.

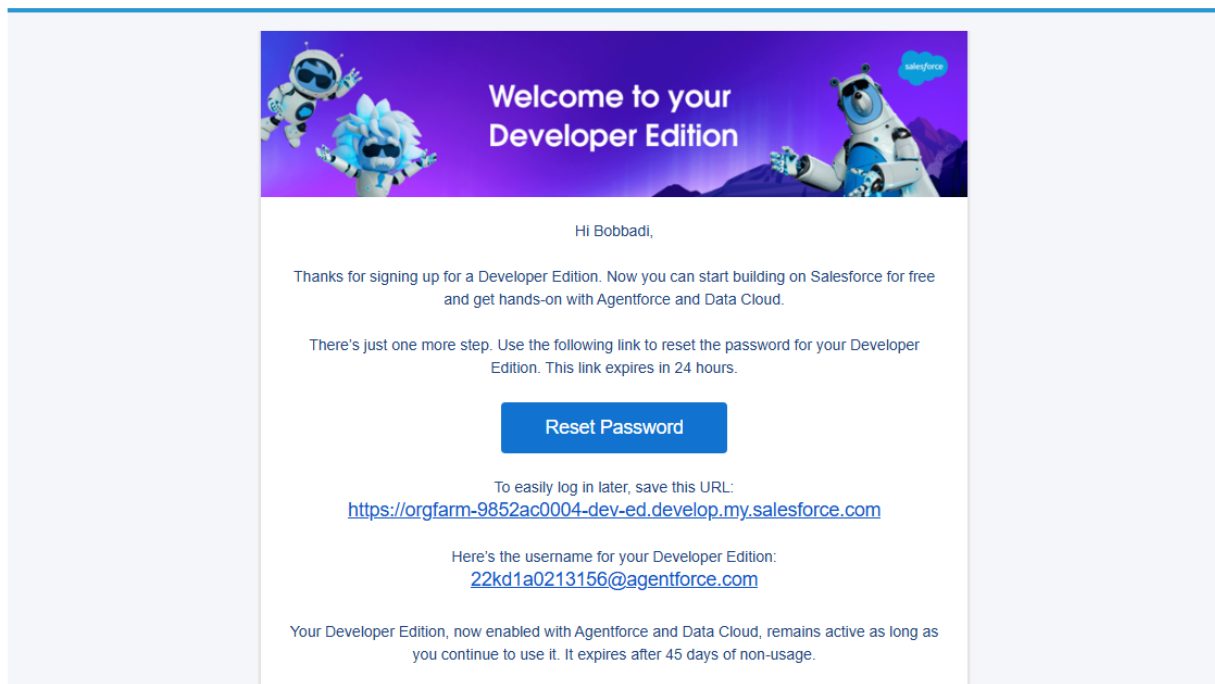
☒ I agree to the Main Services Agreement - Developer Services and Salesforce Program Agreement. I acknowledge, as described in the Developer Documentation: (1) the Developer Edition includes autonomous and other generative AI features; and (2) Salesforce may limit use of those features and the org, and may terminate any org that has been inactive for 45 days.

We value your privacy. To learn more, visit our [Privacy Statement](#).

☐ I'm not a robot 

Activating Salesforce Developer Account

1. Open the inbox of the registered email and verify the account (the email may take 5–10 minutes).
2. Click the Verify Account link.
3. Set a password and security question, then click Change Password.
4. Salesforce will redirect you to the Setup page.



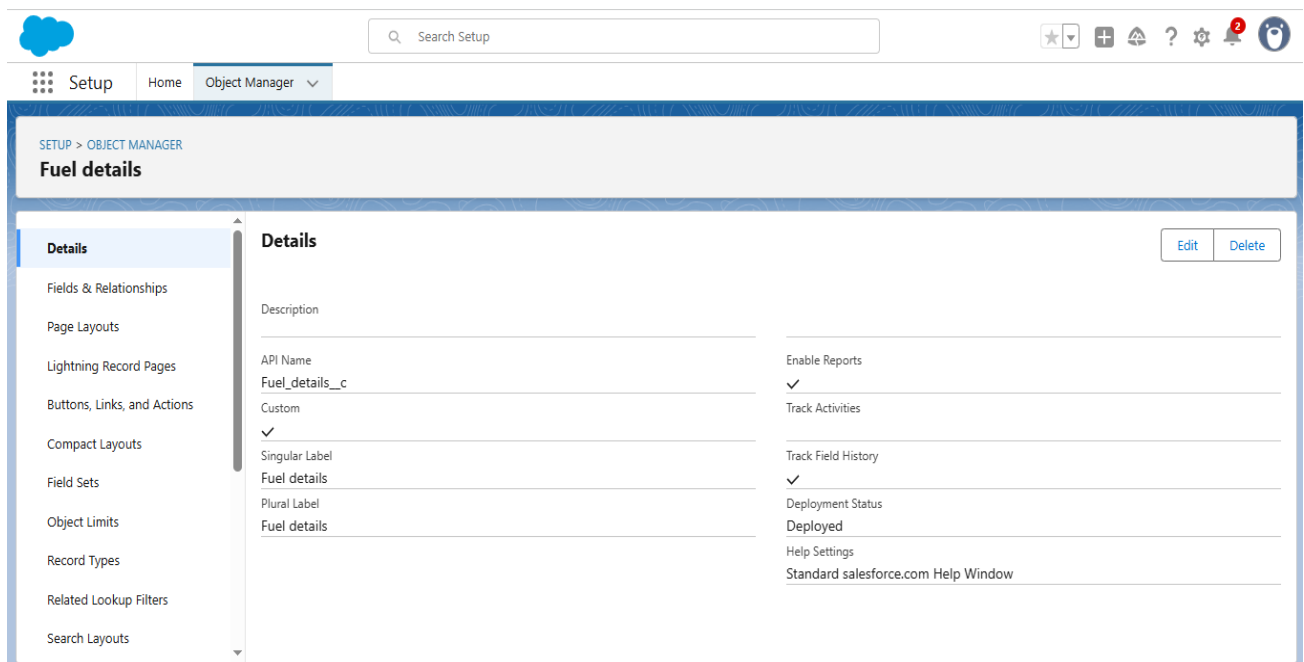
Procedure for Object Creation:

Custom objects in Salesforce are created by defining object names with a __c suffix (e.g., Buyer__c), selecting appropriate data types (e.g., Text, Currency, Date) for fields, and enabling reporting and field tracking for accurate data logging and analysis.

Four Custom Objects:

- Buyer__c: Stores customer details and fuel purchase history.

- Fuel_details__c: Records fuel transactions linking buyers, suppliers, and gas stations.
- Gas_Station__c: Maintains gas station info, including fuel availability and stock levels.
- Supplier__c: Tracks suppliers and the fuel delivered to stations.

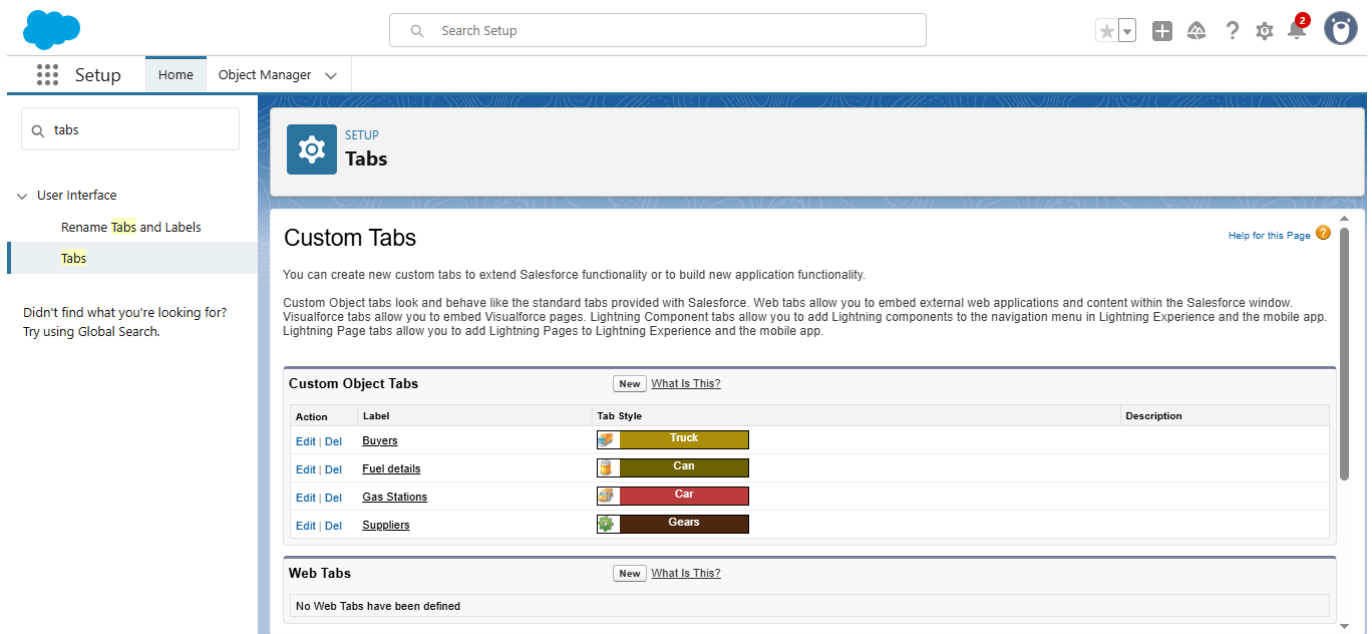


Custom Tabs:

Custom Tabs in Salesforce provide an easy way for users to access and interact with objects. By creating tabs for custom objects, users can directly navigate to the data and manage records without needing to search manually.

In this project, the team created tabs for the following custom objects:

- 1.buyer
- 2.Fuel Details
- 3.Gas Station
- 4.Supplier



Create Lightning App:

A Lightning App in Salesforce allows grouping of related objects, tabs, and functionalities into a single workspace for easy access and management.

For this project, the team created a dedicated app named 'GAS STATION' to handle CRM operations effectively.

Key steps:

- Created the Lightning App “GAS STATION”.
- Assigned the app to the System Administrator profile for access and management.
- Configured the app with the required custom objects:

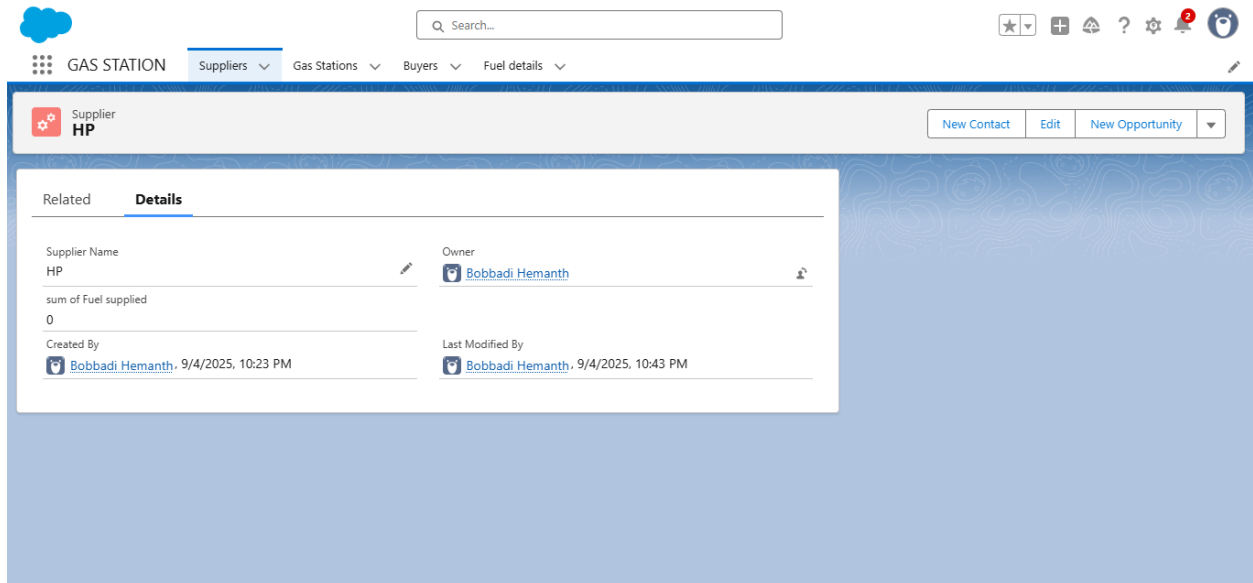
1.Buyer1

2.Fuel Details

3.Gas Station

4.Supplier

- Added the necessary fields in each object to ensure smooth operations of the Gas Station CRM.
- This setup allows users to manage all records and processes related to the gas station from a single, user-friendly application within Salesforce.



Fields to be created:

- Buyer__c: First Name, Last Name, Customer Name (Formula), Phone Number, Email, Vehicle Type (Picklist), Fuel Filled in Vehicle, Mode of Payment (Picklist), Amount Paid (Formula)
Validation in buyer object (phone number)

The screenshot shows the Salesforce Setup interface. At the top, there's a navigation bar with 'Setup', 'Home', and 'Object Manager'. Below this, the 'Buyer' object is selected. The left sidebar lists various setup options like 'Details', 'Fields & Relationships', 'Page Layouts', etc. The main content area displays the 'Buyer Validation Rule' configuration. The rule is named 'Phone' and is currently active. The error condition formula is 'NOT(REGEX(Phone_number__c, "[6-9]{1}[0-9]{9}"))'. The error message is 'incorrect data' and the error location is 'Phone number'. The rule was created and modified by 'Bobbadi Hemanth' on 9/3/2025 at 3:09 AM.

SETUP > OBJECT MANAGER
Buyer

Details
Fields & Relationships
Page Layouts
Lightning Record Pages
Buttons, Links, and Actions
Compact Layouts
Field Sets
Object Limits
Record Types
Related Lookup Filters
Search Layouts

Buyer Validation Rule

[Back to Buyer](#)

Validation Rule Detail

Rule Name	Phone	Active	✓
Error Condition Formula	NOT(REGEX(Phone_number__c, "[6-9]{1}[0-9]{9}"))		
Error Message	incorrect data	Error Location	Phone number
Description			
Created By	Bobbadi Hemanth, 9/3/2025, 3:09 AM	Modified By	Bobbadi Hemanth, 9/3/2025, 3:09 AM

Fields to be Created:

Fuel_Details__c:

- Fuel Supplied (Number)
- Supplier Name (Master-Detail)
- Gas Station (Master-Detail)

Gas_Station__c:

- Fuel Price per Liter (Number)
- Fuel Supplied to Bunk (Roll-up Summary)
- Fuel Used (Roll-up Summary)
- Fuel Available in Bunk (Formula)

Supplier__c:

- Sum of Fuel Supplied (Roll-up Summary)

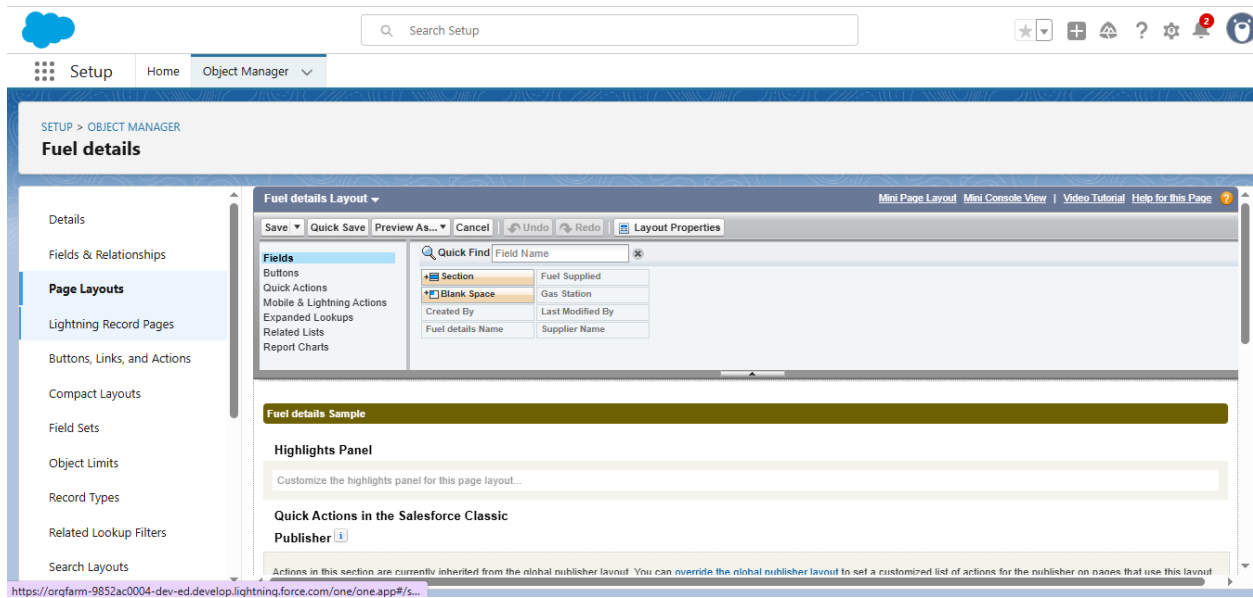
The screenshot shows the Salesforce Setup interface. At the top, there's a search bar labeled 'Search Setup'. Below it, the navigation menu includes 'Setup', 'Home', and 'Object Manager'. The 'Object Manager' section is expanded, showing 'Supplier' as the selected object. On the left, a sidebar lists various setup options: Details, Fields & Relationships (selected), Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, Record Types, Related Lookup Filters, and Search Layouts. The main content area displays the 'Fields & Relationships' section for the 'Supplier' object, showing 5 items sorted by Field Label. A table lists the fields with columns for Field Label, Field Name, Data Type, Controlling Field, and Indexed status.

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedById	Lookup(User)		
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)		✓
sum of Fuel supplied	sum_of_Fuel_supplied__c	Roll-Up Summary (SUM Fuel details)		
Supplier Name	Name	Text(80)		✓

Create Page Layout for Objects:

The team designed page layouts for each custom object to structure sections and position fields for easy access and a better user experience

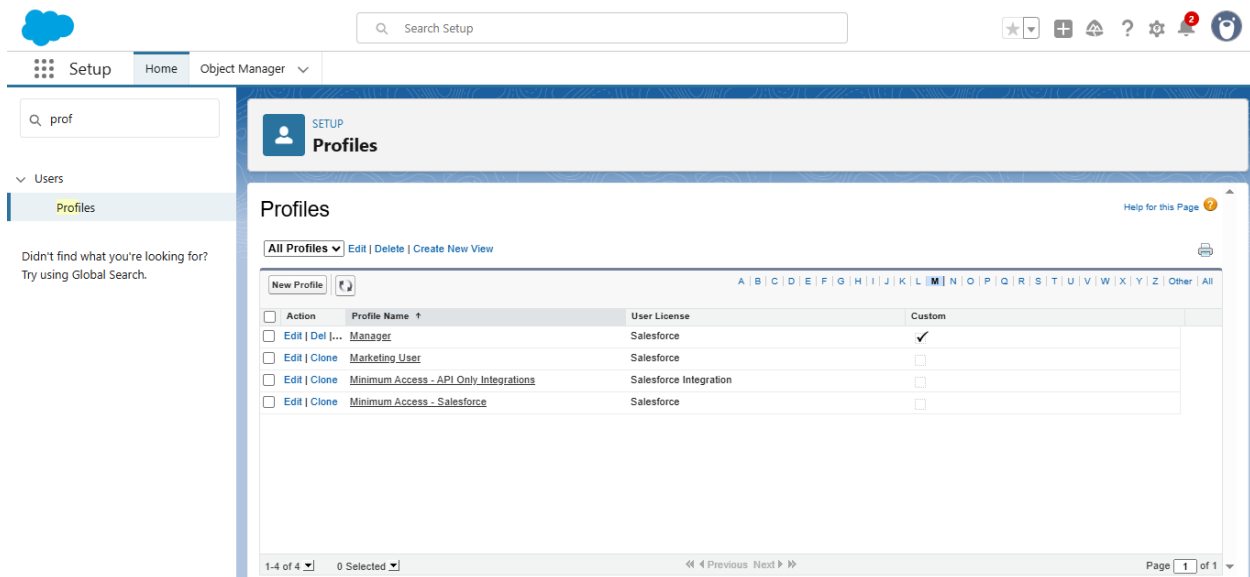
- Navigate to Setup → Object Manager → [Object] → Page Layouts → New Layout.
- Add sections as required and place the corresponding fields within them.
- Reorder fields logically to ensure a clean and user-friendly layout.
- Click Save to activate the layout for the selected object.



Create Profiles:

The team created and customized profiles to manage user permissions and control access to objects, fields, and applications..

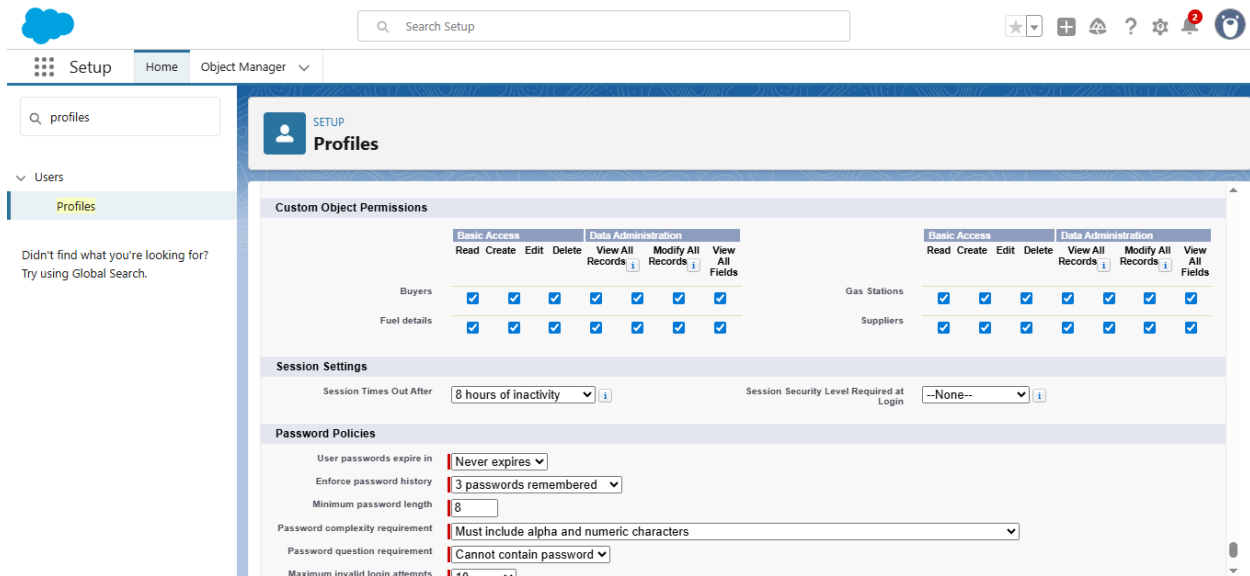
- Go to Setup → Profiles → New Profile.
- Select an existing profile as a baseline (e.g., Standard User or System Administrator).
- Define object permissions, field-level security, and tab visibility as per requirements.
- Assign the profile to users for appropriate access levels.
- Save the profile settings.



The screenshot shows the Salesforce Setup interface. The left sidebar has a search bar with "prof" entered and a "Profiles" link under the "Users" section. The main content area is titled "Profiles" and shows a list of profiles. The "Manager" profile is selected, and its "Custom" checkbox is checked. The table below shows the list of profiles.

Action	Profile Name	User License	Custom
Edit Delete Clone	Manager	Salesforce	☒
Edit Delete Clone	Marketing User	Salesforce	☐
Edit Delete Clone	Minimum Access - API Only Integrations	Salesforce Integration	☐
Edit Delete Clone	Minimum Access - Salesforce	Salesforce	☐

- Click Edit to customize the app settings and set object permissions for the relevant objects.
- Modify Session Settings and Password Policies according to the project requirements.



The screenshot shows the Salesforce Setup interface for customizing a profile. The left sidebar has a search bar with "profiles" entered and a "Profiles" link under the "Users" section. The main content area is titled "Profiles" and shows the "Custom Object Permissions" section. The "Manager" profile is selected, and its "Custom" checkbox is checked. The table below shows the list of profiles.

	Basic Access				Data Administration				Basic Access				Data Administration			
	Read	Create	Edit	Delete	View All Records	Modify All Records	View All Fields		Read	Create	Edit	Delete	View All Records	Modify All Records	View All Fields	
Buyers	☒	☒	☒	☒	☒	☒	☒		☒	☒	☒	☒	☒	☒	☒	
Fuel details	☒	☒	☒	☒	☒	☒	☒		☒	☒	☒	☒	☒	☒	☒	
Gas Stations	☒	☒	☒	☒	☒	☒	☒		☒	☒	☒	☒	☒	☒	☒	
Suppliers	☒	☒	☒	☒	☒	☒	☒		☒	☒	☒	☒	☒	☒	☒	

Session Settings

Session Times Out After: 8 hours of inactivity

Session Security Level Required at Login: --None--

Password Policies

User passwords expire in: Never expires

Enforce password history: 3 passwords remembered

Minimum password length: 8

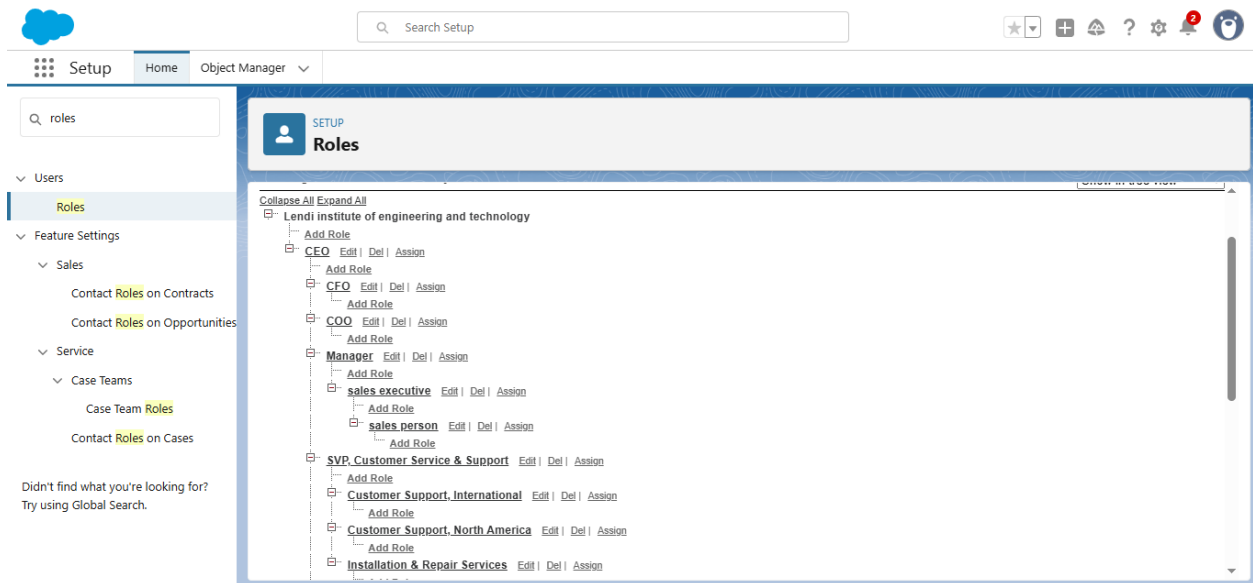
Password complexity requirement: Must include alpha and numeric characters

Password question requirement: Cannot contain password

Maximum invalid login attempts: 5

Creating Roles & Hierarchy:

- Navigate to Setup → Roles → Set Up Roles, then click Expand All. Click Add Role under the appropriate parent role.
- Enter the Role Label (e.g., Manager); the Role Name will auto-fill. Click Save.
- Add Sales Executive and Sales Person roles under the Manager and assign the respective profiles.
- Make sure the hierarchy represents the reporting structure: Manager → Sales Executive → Sales Person.



Creating Users:

- Users in Salesforce are accounts that provide access based on assigned roles and profiles.
- For this project, the team created users for different roles:
 - Exec Sales – Sales Executive
 - Michelson, Nicklaus – Manager
 - Person, Sales – Sales Person

The screenshot shows the 'User Edit' page for Niklaus Mikaelson. The left sidebar contains navigation links: Setup, Home, Object Manager, Users, Permission Set Groups, Permission Sets, Profiles, Public Groups, Queues, Roles, User Management Settings, Feature Settings, Data.com, and Prospector Users. The main content area is titled 'User Edit' and includes a 'General Information' section with fields for First Name, Last Name, Alias, Email, Username, Nickname, Title, Company, Department, and Division. The 'Role' is set to 'Manager', 'User License' is 'Salesforce', and 'Profile' is 'Manager'. The 'Active' checkbox is checked. There are also checkboxes for Marketing User, Offline User, Knowledge User, Flow User, Service Cloud User, Site.com Contributor User, Site.com Publisher User, and WDC User. Buttons for 'Save', 'Save & New', and 'Cancel' are at the top right of the form.

Created users the displayed

The screenshot shows the 'All Users' page in Salesforce Setup. It displays a table of users with columns for Action, Full Name, Alias, Username, Role, Active, and Profile. The table lists several users, including Chatter Expert, EPIC_OrgFarm, Hemanth Bobbadi, Mikaelson Niklaus, mukesh chitti, rohith dhomma, User Integration, and User Security. Buttons for 'New User', 'Reset Password(s)', and 'Add Multiple Users' are at the top of the table. The left sidebar is the same as in the previous screenshot.

Action	Full Name	Alias	Username	Role	Active	Profile
Edit	Chatter Expert	Chatter	chatty.00d9k00000aigtvuah_ylmzhovrksb@chatter.salesforce.com		✓	Chatter Free User
Edit	EPIC_OrgFarm	OEPIK	epic.0d12c4bee87e@orgfarm.salesforce.com		✓	System Administrator
Edit	Hemanth Bobbadi	22k	22kd1a0213156@agentforce.com		✓	System Administrator
Edit	Mikaelson Niklaus	nmika	hemanth@lendi.edu	Manager	✓	Manager
Edit	mukesh chitti	chitti	mukesh@lendi.edu	sales executive	✓	sales executive
Edit	rohith dhomma	drohit	rohith@lendi.edu	sales person	✓	sales person
Edit	User Integration	integ	integration@00d9k00000aigtvuah.com		✓	Analytics Cloud Integration User
Edit	User Security	sec	insightssecurity@00d9k00000aigtvuah.com		✓	Analytics Cloud Security User

Creating Permission Sets:

- Permission Sets provide additional access to users without modifying their profiles.
- The project includes a permission set that grants extra access to relevant objects.

- The team assigned the P1 Permission Set to the Sales Executive.

The screenshot shows the Salesforce Setup interface. The left sidebar has a search bar with "permis" and a list of categories: Users, Permission Set Groups, Permission Sets (selected), and Custom Code. The main content area shows the "P1" permission set configuration. Under "Current Assignments", there is a table with columns: Full Name, Active, Role, Profile, User License, and Expires On. One assignment is listed for "chitti mukesh" with the role "sales executive" and profile "sales executive" on the "Salesforce Platform" license.

Full Name ↑	Active	Role	Profile	User License	Expires On
chitti mukesh	✓	sales executive	sales executive	Salesforce Platform	

Organization-Wide Defaults (OWD):

- OWD settings determine the default access users have to records they do not own.
- In this project:
 - The team set the Gas Station and Supplier objects to Public Read-Only.
 - The team managed additional access through roles and profiles.

The screenshot shows the "Sharing Settings" page in Salesforce Setup. The left sidebar has a search bar with "sharin" and a list of categories: Security, Guest User Sharing Rule Access Report, and Sharing Settings (selected). The main content area shows a table of sharing settings for various objects. The table has columns for the object name, a dropdown menu for the sharing setting, and a checkbox for "Public Read Only".

Object	Sharing Setting	Public Read Only
Tableau most mapping	Public Read Only	✓
User Presence	Public Read Only	✓
Waitlist	Private	✓
Web Cart Document	Private	✓
Work Order	Private	✓
Work Plan	Private	✓
Work Plan Template	Private	✓
Work Step Template	Private	✓
Work Type	Private	✓
Work Type Group	Public Read/Write	✓
Gas Station	Public Read Only	✓
Supplier	Public Read Only	✓

Other Settings:

- Standard Report Visibility: ☒ *i*
- Manual User Record Sharing: ☐ *i*
- Manager Groups: ☐ *i*
- Secure guest user: ☐ *i*
- Require permission to view record access: ☒ *i*
- record names in lookup fields: ☐ *i*

Buttons: Save, Cancel

Create, View, Delete Records:

- User adoption reflects how effectively users interact with Salesforce features.
- In this project, implementing roles, profiles, permission sets, and intuitive page layouts helped ensure smooth usage and increased adoption of the Gas Station CRM application.

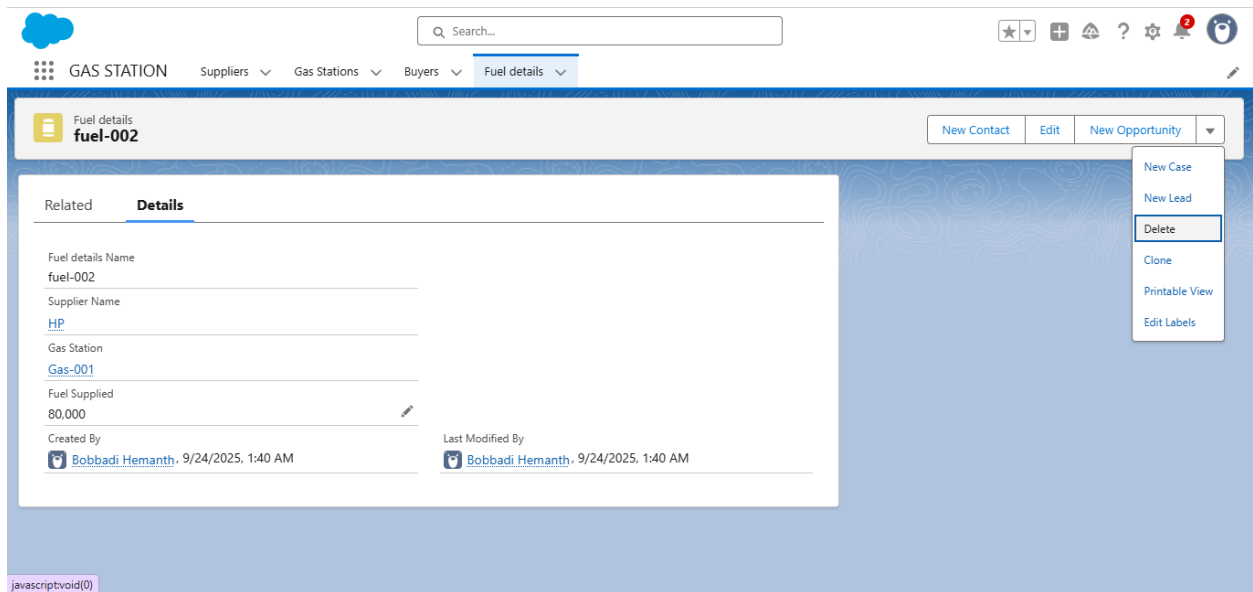
• Create record:

The screenshot shows a 'New Fuel details' modal form. The form has a title bar with a close button. Below the title bar, there's a section for 'Information' with a legend indicating that an asterisk (*) denotes required information. The form contains three required fields: 'Supplier Name' (with a dropdown menu showing 'HP'), 'Gas Station' (with a dropdown menu showing 'Gas-001'), and 'Fuel Supplied' (with a text input field containing '80000'). At the bottom of the modal, there are three buttons: 'Cancel', 'Save & New', and 'Save'. The background shows a sidebar with 'GAS STATION' and 'Suppliers' tabs, and a main area with a 'Recently Viewed' section.

• View record:

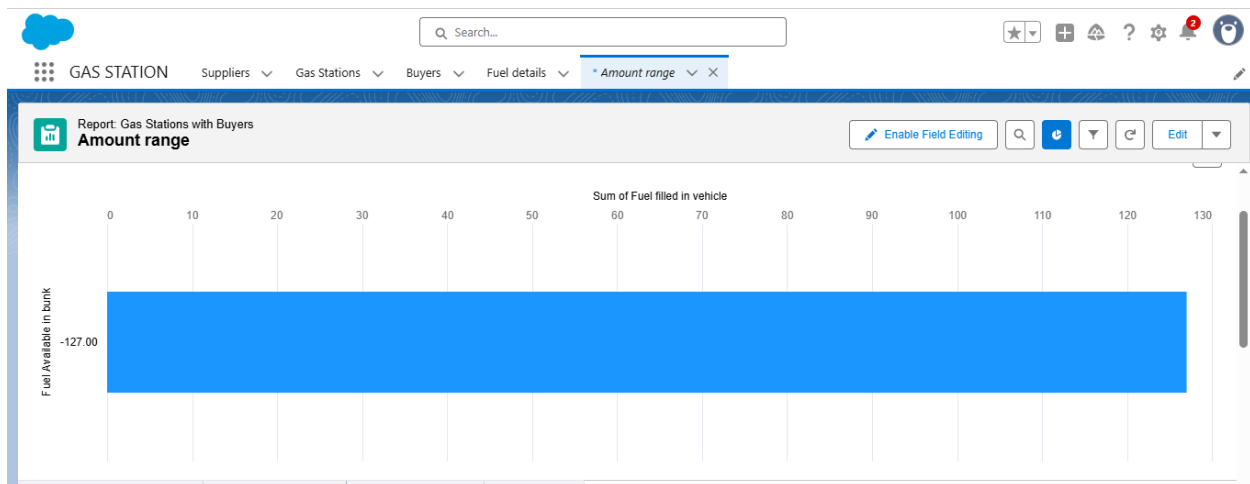
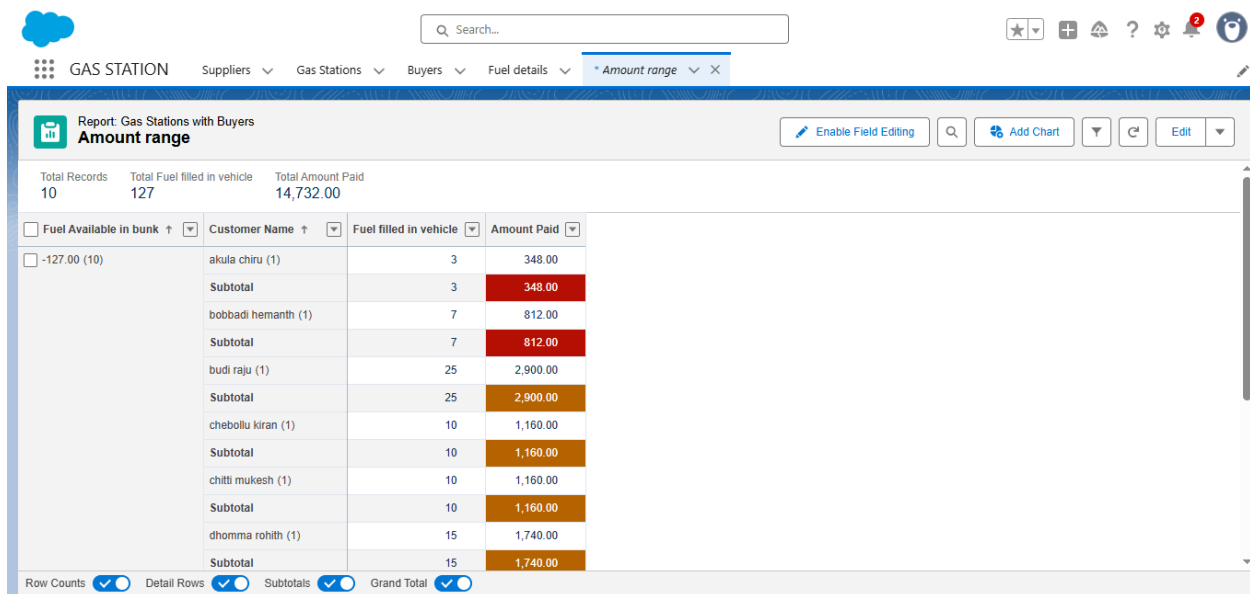
The screenshot shows the 'Details' view for a 'Supplier' record named 'HP'. The record is owned by 'Bobbadi Hemanth'. The details section shows the following information: 'Supplier Name' is 'HP', 'sum of Fuel supplied' is '0', 'Created By' is 'Bobbadi Hemanth' on '9/4/2025, 10:23 PM', and 'Last Modified By' is 'Bobbadi Hemanth' on '9/4/2025, 10:43 PM'. The background shows a sidebar with 'Supplier' and 'HP' tabs, and a main area with a 'Related' section.

- **Delete record:**



Creating Reports:

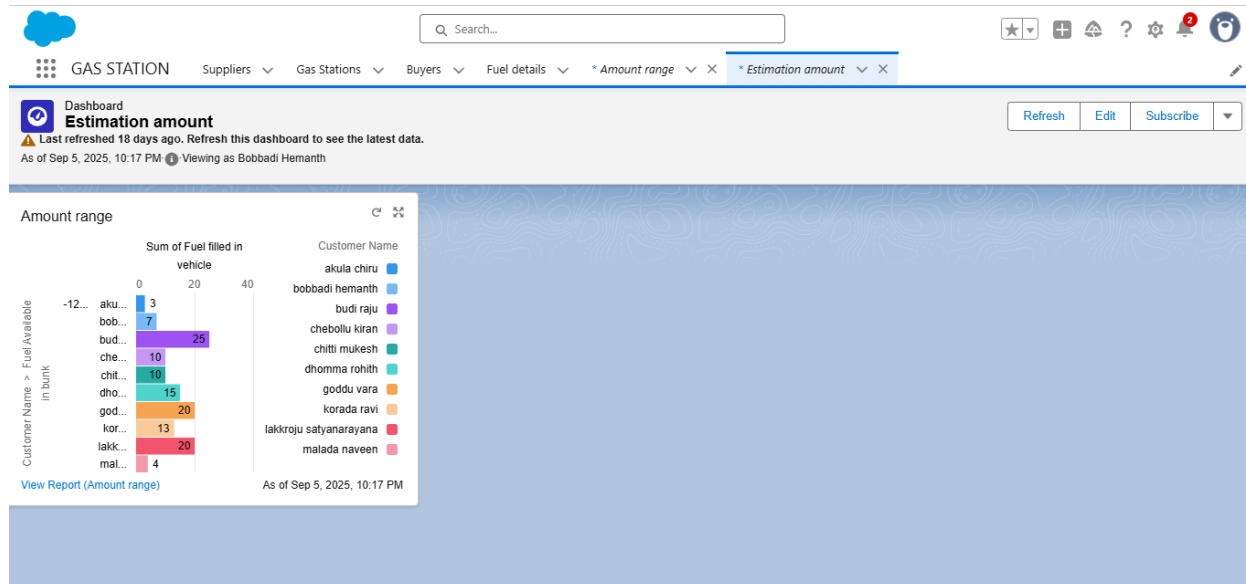
- Reports help organize and present Salesforce data in a clear, structured way.
- Custom reports monitor fuel usage, customer activity, and sales, enabling more informed decision-making.



Reports and Dashboards:

- Reports allow tracking and analyzing Salesforce data.
- Dashboards display report data visually for better insights.
- In this project:
 - The team created reports for fuel sales, supplier performance, and inventory levels.
 - The team configured dashboards to display charts and graphs for Managers and

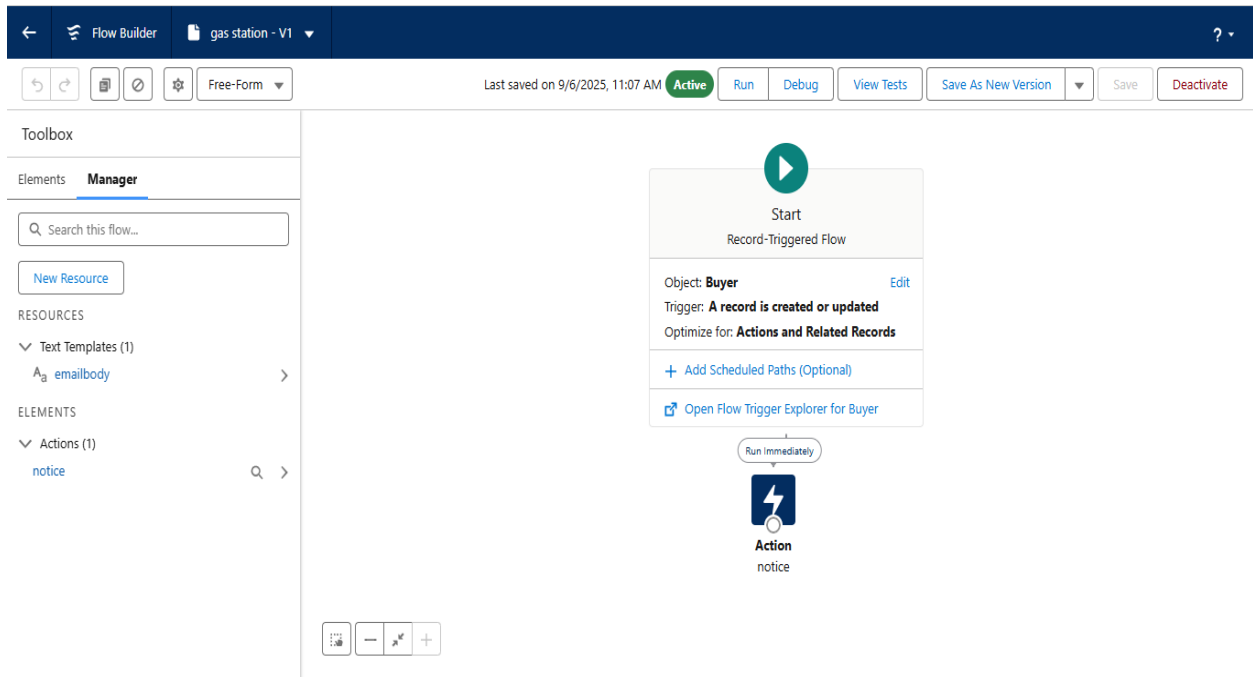
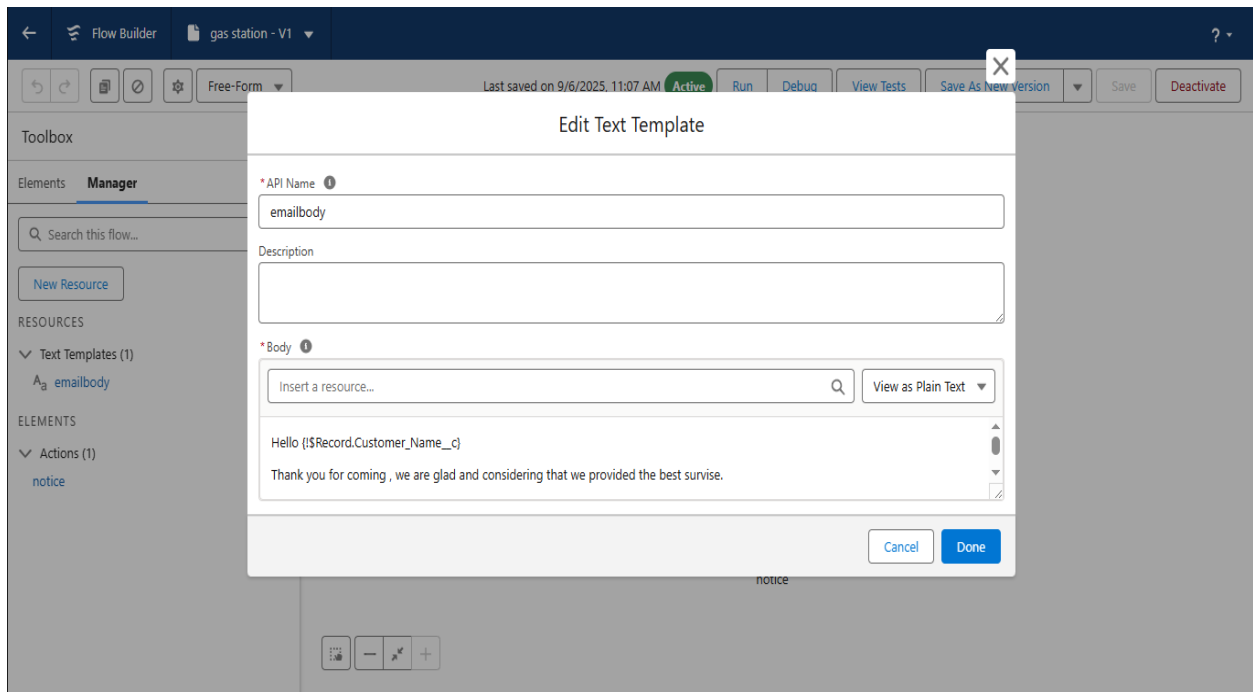
Executives.



Creating Flows:

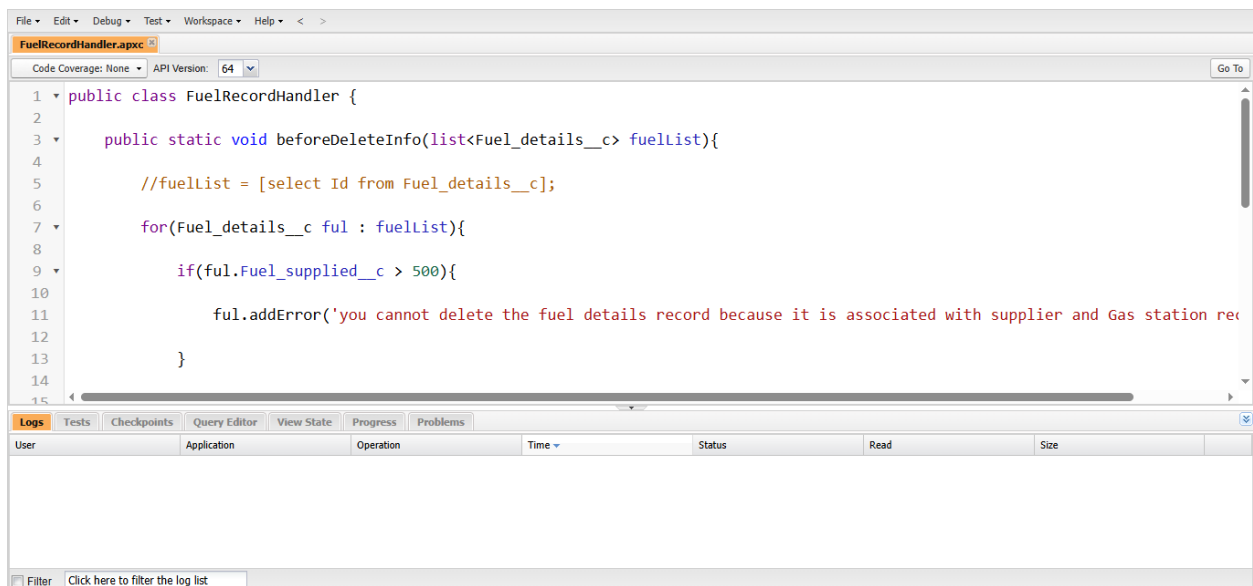
Flows help reduce manual effort and ensure faster communication with customers.

- In this project, the team created a record-triggered flow on the Buyer object.
- The flow triggers on record creation or update.
- It automatically sends an email receipt to the customer.
- The email includes Customer Name, Amount Paid, Vehicle Type, and Fuel Intake.



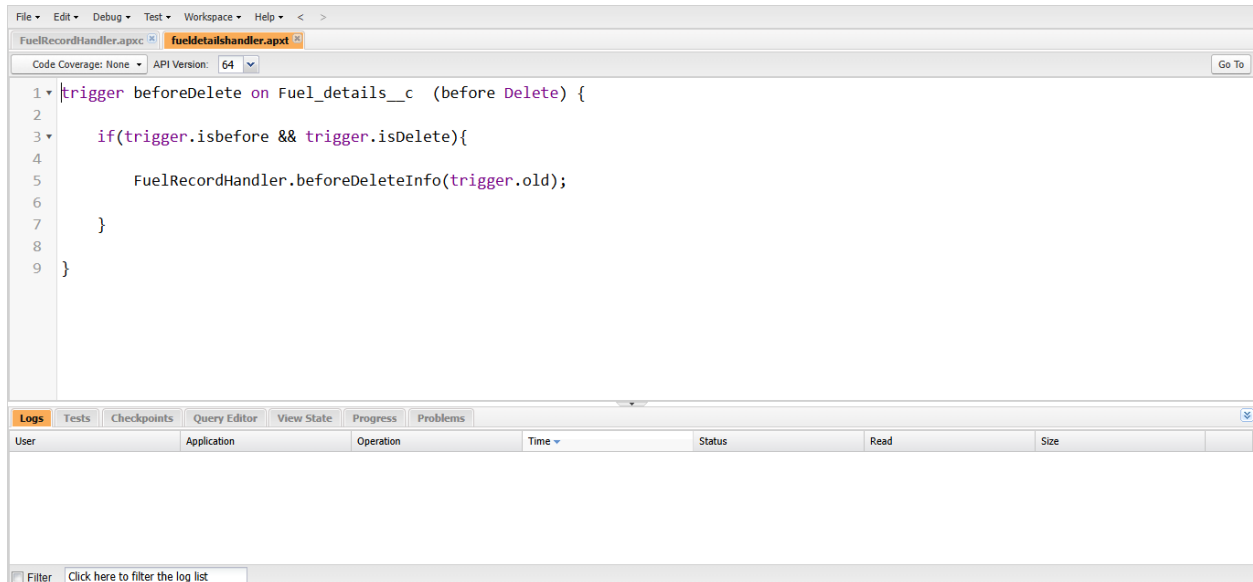
Creating Apex Triggers:

- Apex triggers are custom pieces of code that execute before or after events on records.
- In this project, a handler (FuelRecordHandler.apxc) was used with the following triggers:
 - beforeDeleteInfo – Prevents deletion of Fuel Details if the supplier quantity is greater than 500.
 - beforeDeleteGas – Ensures the Gas Station fuel price is above 50 before inserting a record.



Trigger: Fuel Details (before delete)

Stops deletion of Fuel Details when associated records exist.



Trigger: Gas Station (before insert)

Checks and validates the fuel price before a record is created.



Smart Automation & Predictive Insights:

The team can integrate Salesforce Einstein features into the Gas Station app to enhance customer experience, improve business decision-making, and automate key processes:

Einstein Bots Integration – AI-powered chatbots can handle customer queries, provide fuel prices, and generate receipts automatically.

Einstein Next Best Action – Offers personalized discounts or promotions based on customers' fuel purchase history.

Einstein Activity Capture – Automatically logs emails and communications with customers for better tracking and engagement.

Einstein Analytics Dashboard – Visualizes sales trends, supplier performance, and customer demand using advanced analytics.

Einstein Prediction Builder – Predicts customer return behavior to enable targeted marketing campaigns.

Einstein Recommendation Builder – Suggests fuel or service options to customers based on preferences and past behavior.

Einstein Opportunity Insights – Provides alerts and recommendations to increase sales.

Conclusion:

The Gas Station CRM project in Salesforce has been successfully designed and implemented to streamline all key business operations, including managing buyers, suppliers, fuel inventory, and sales. By creating custom objects, profiles, permission sets, and page layouts, the application ensures organized data management and secure access for users such as Managers, Sales Executives, and Sales Persons. Automated flows improve efficiency by sending customer receipts and reducing manual tasks, while Apex triggers maintain data integrity during record creation and deletion. The use of reports and dashboards provides actionable insights, enabling managers to monitor performance, analyze trends, and make informed decisions. Furthermore, integrating Salesforce Einstein features enhances customer experience through AI-powered chatbots, predictive insights, and personalized recommendations. Overall, this project shows how the team leveraged Salesforce to increase operational efficiency, improve data accuracy, and deliver a superior customer experience in gas station management.